



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION, SEMESTER II – October, 2016
ST102 – INTRODUCTION TO PROBABILITY THEORY – PART I.

Time Allowed: **TWO Hours**

Answer **All** Questions.

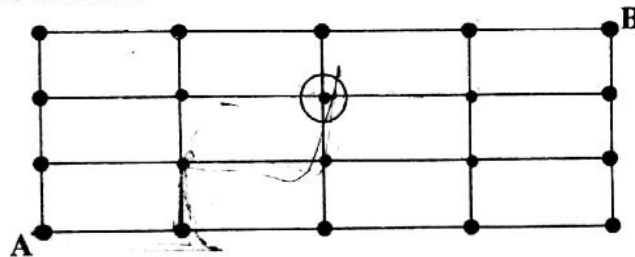
1.

a). Show that

$$k \binom{n}{k} = (n - k + 1) \binom{n}{k-1} = n \binom{n-1}{k-1}$$

b). Consider a set of n antennas of which m are defective. Suppose that all of the defectives and all of the non defective antennas are considered indistinguishable. How many linear orderings are there in which no two defectives are consecutive?

c). Consider the grid of points shown below. Suppose that starting from point "A", you can go one step up or one step to the right at each move. This is continued until the point "B" is reached.



What is the probability that the chosen path contains the point circled in the figure?

2. Let X be a continuous random variable with the following probability density function (pdf).

$$f(x) = \begin{cases} \frac{x}{25} & ; 0 \leq x < 5 \\ \frac{2}{5} - \frac{x}{25} & ; 5 \leq x < 10 \\ 0 & ; \text{o/w} \end{cases}$$

- Show that $f(x)$ is a legitimate pdf.
- Find $E(X)$ and $Var(X)$.
- Derive the Cumulative Density Function (CDF) of X .
- Calculate $P(3 < x < 6)$.
- Calculate $P(5 < x < 8 | x > 7)$.

[P.T.O.]

3.

a). Suppose X has a binomial (n, p) distribution and let Y have a negative binomial (r, p) distribution. Show that $P[X < (r - 1)] = 1 - P[Y < (n - r)]$.

b). Two movie theaters compete for the business of 1000 customers. Assume that each customer chooses between the movie theaters independently and with equally likely. Let N denote the number of seats in each theater.

(i). Using a binomial model, find an expression for N that will guarantee that the probability of turning away a customer (because all seats are full) is less than 1%.

(ii). Use the normal approximation to get a numerical value for N .

4. Let a continuous random variable X follows uniform distribution within the interval $[\alpha, \beta]$.

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & ; \alpha \leq x \leq \beta \\ 0 & ; \text{o/w} \end{cases}$$

a). Derive the Moment Generating Function (MGF) of X .

b). Derive an expression for the k^{th} moment about the origin.

c). Show that the mean and median are same for X .

d). You arrived at a bus stop at 10'o clock, knowing that the bus will arrive at some time uniformly distributed between 10 and 10:30. If at 10:10 the bus has not yet arrived, what is the probability that you will have to wait at least an additional 15 minutes?

Distribution	Density Function
Binomial (n, θ)	$P(X = x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}; x = 0, 1, 2, \dots, n$
Negative Binomial (k, θ)	$P(X = x) = \binom{x-1}{k-1} \theta^k (1 - \theta)^{x-k}; x = k, k+1, k+2, \dots$
Geometric (θ)	$P(X = x) = \theta^x (1 - \theta)^{x-1}; x = 1, 2, \dots$

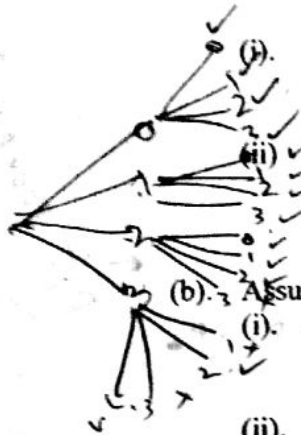


FACULTY OF SCIENCE - UNIVERSITY OF PERADENIYA
END SEMESTER EXAMINATION - SEMESTER II, 2011(SEPT)
INTRODUCTION TO PROBABILITY THEORY - ST102/MT102 (3 CREDITS)

Time Allowed: Two hours
 Calculators will be provided

Answer All questions

1. (a). An operation consists of two steps, of which the first can be made in n_1 ways. If the first step is made in the i th way, the second step can be made in n_{2i} ways.



(i). Use a tree diagram to find a formula for the total number of ways in which the total operations can be made.

(ii). A student can study 0, 1, 2 or 3 hours for a *Statistics* test on any given day. Use the formula obtained in part (i) to verify that there are 13 ways in which the student can study at most 4 hours for the test on two consecutive days.

Assume that A , B , and C are any two events in a sample space S such that $P(A \cap B) \neq 0$.

(i). Assuming that the three events are NOT independent, clearly write an expression to calculate the probability $P(A \cap B \cap C)$.

(ii). Prove the expression given in (b-i).

(iii). By using the proof given in (b-ii) derive an expression to calculate the probability $P(A \cap B \cap C)$ assuming that the three events are independent.

(iv). A coin is tossed three times and the eight possible outcomes, HHH, HHT, HTH, THH, HTT, THT, TTH, and TTT, are assumed to be equally likely. If A is the event that a head occurs on each of the first two tosses, B is the event that a tail occurs on the third toss, and C is the event that exactly two tails occur in the three tosses, show that,

- (p). events A and B are independent;
 (q). events B and C are dependent.

2. (a). (i). Based on the three axioms of probability, clearly state, the two conditions that should be satisfied by a probability mass function. $f(x)$

(ii). For the function, $f(x) = c \left(\frac{5}{x}\right)$, determine c so that the function can serve as $f(x)$ probability distribution function of the random variable X . ($x = 1, 2, 3, 4, 5$)

(iii). Find the distribution function of the random variable X (given in 2-a-ii). $F(x)$

(b). The probability density of a random variable Y is given by

$$f(y) = \begin{cases} \frac{1}{8}(y+1), & \text{for } 2 < y < 4 \\ 0 & \text{elsewhere} \end{cases}$$

Find $P(Y < 3.2)$ and $P(2.9 < Y < 3.2)$

[P.T.O]

[Continued]

2. [Continued]

- (c) The amount of time it takes a person to be served at a given restaurant is a random variable with the probability density

$$f(x) = \begin{cases} \frac{1}{4} e^{-\frac{x}{4}} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Find the mean and the variance of this random variable.

3. Identify suitable distribution (or density) functions for the following scenarios, and then give suitable solutions for the given questions.

- (a) Of the people passing through an airport metal detector, 5% activate it; let X be the number among a randomly selected group of 500 who activate the detector.

- (i) What is the *probability mass function* of X ? What is/are the value(s) of parameter(s) of the distribution?

- (ii) What is the probability that exactly 5 people will be detected by metal detector?

- (iii) What is the probability that at most 6 people will be detected by the metal detector?

$$P_0 + P_1 + P_2 + P_3 + P_4 + P_5 + P_6$$

- (b) When circuit boards used in manufacture of compact disc (CD) players are tested, the long-run percentage of defectives is 5%. Assume that there are 25 boards in a sample.

- (i) What is the *probability mass function* of X (number of defective boards)? What is/are the value(s) of parameter(s) of the distribution?

- (ii) Determine the probability that the number of defective boards in the sample is between 1 and 4 (inclusive of both numbers).

- (iii) Calculate the expected value and standard deviation of X .

Binomial

- (c) If 18 defective glass bricks include 10 that have cracks but no discoloration, five that have discoloration but no cracks, and three that have cracks and discoloration, what is the probability that among six of the bricks (chosen at random for further check) three will have cracks but no discolorations, one will have discoloration but no cracks, and two will have cracks and discoloration?

- (d) Suppose that calls are received at "emergency hot line" (119) about suicides with the rate of 0.5 calls per day.

- (i) What is the probability that at least two calls received on a given day?

- (ii) What is the probability that more than two days elapse between two such successive calls?

$$(2, 0.5) \rightarrow 1 - \{P(0, 1) + P(1, 1)\}$$

4. (a) Using moment-generating function of the binomial distribution, $M_X(t) = [1 + \theta(e^t - 1)]^n$, derive the mean and the variance of the binomial distribution.

- (b) Suppose the force acting on a column (concrete post) that helps to support a building is normally distributed with mean 15.0 kips (a unit measuring the strength) and standard deviation 1.25 kips. What is the probability that the force

- (i) Is at most 17 kips?

- (ii) Is between 10 and 12 kips?

- (iii) Differs from 15.0 kips by at most 2 standard deviations?

- (c) If the probability is 0.20 that a certain bank will refuse a loan application, using a suitable approximation determine the probability that the bank will refuse at most 40 of 225 loan applications.



FACULTY OF SCIENCE - UNIVERSITY OF PERADENIYA

END SEMESTER EXAMINATION - SEMESTER II, 2007

INTRODUCTION TO PROBABILITY THEORY ST 102/MT 102 (3 credits)

Allowed: two hours.
Calculators will be provided.

Answer ALL questions.

(i) Two events A and B are such that $P(A) = 8/15$, $P(B) = 1/3$ and $P(A|B) = 1/5$.

Calculate the probability that

- (a) both events occur (b) only one of the two events occur.

(ii) A and B are about to play a series of games of table tennis. From past records it is known that A has the probability 0.6 of winning the first game. In the other games, A's probability of winning a game is 0.7 if he has won the previous game but is only 0.5 if he has lost the previous game. (There are no draws.) Find the probability that A will win the third game in the series. (A tree diagram may be helpful.)

The continuous random variable X has the p.d.f. given by

$$f(x) = \begin{cases} cx^2; & 0 \leq x \leq 2 \\ 2c(4-x); & 2 \leq x \leq 4 \\ 0; & \text{otherwise.} \end{cases} \quad \text{where } c \text{ is a constant.}$$

- a) Show that $c = 0.15$ (You must show your work.)
b) Find the mean of X.
c) Find the lower quartile of X (That is, find q_1 such that $P(X \leq q_1) = 0.25$).
d) Find the probability that a single observation of X lies between the lower quartile and the mean.

If $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$ are independent random variables, what is the distribution of $aX + bY$ when a and b are constants.

The diameters of apples grown in an orchard are normally distributed with mean 8.5 cm and standard deviation 1 cm.

- a) What proportion of apples have diameters greater than 10 cm? less than 7.5 cm?
b) If two apples are selected at random, what is the probability that the difference in their diameters is more than 1.25 cm?
c) If four apples are selected at random, what is the probability that the sum of the diameters is more than 40 cm?

4. (a) Let $f(x) = \begin{cases} 3x & \text{if } x \text{ is rational,} \\ 2x+1 & \text{if } x \text{ is irrational.} \end{cases}$

Let $a \in \mathbb{R}$. Also let (r_n) and (s_n) be sequences of rational numbers and irrational numbers respectively, such that $\lim_{n \rightarrow \infty} r_n = \lim_{n \rightarrow \infty} s_n = a$.

Find $\lim_{n \rightarrow \infty} f(r_n)$ and $\lim_{n \rightarrow \infty} f(s_n)$ and deduce that if $a \neq 1$, then f is not continuous at a .

Find a value for the constant $k (> 0)$ such that $|f(x) - f(1)| \leq k|x - 1|$ for all $x \in \mathbb{R}$.

Deduce that f is continuous at 1.

- (b) Let $f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$

Show that $f'(0) = 0$.

5. (a) State Rolle's Theorem.

Verify Rolle's Theorem for the function $f: [a, b] \rightarrow \mathbb{R}$ given by $f(x) = (x-a)^3(x-b)^2$.

- (b) State and prove the Mean-Value Theorem.

Using the Mean-Value Theorem, show that $\sqrt{1+x} < 4 + \frac{1}{8}(x-15)$ for $x > 15$.

6. Find each of the following limits:

(i) $\lim_{x \rightarrow 0} \frac{\sin x - x}{e^x - 1}$

(ii) $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\tan x} \right)$

(iii) $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$

(iv) $\lim_{n \rightarrow \infty} (2^n + 3^n)^{\frac{1}{n}}$



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
ST/MT 102 – INTRODUCTION TO PROBABILITY THEORY
MID SEMESTER EXAMINATION - AUGUST 2015

Time Allowed: **ONE hour**

Answer **All** Questions.

1. a) A stack contains 10 text books. There are 5 Statistics books, 3 Mathematics books and 2 Physics books. If you are allowed to choose only 5 books out of the 10, what is the probability that you choose
- 3 Statistics books and 2 Mathematics books.
 - At least one from each subject.
 - More Mathematics books than Statistics books.
- b) In a group of 400 citizens, 175 are males. Out of them 100 suffer from diabetes. Among the females, 200 suffer from diabetes. If a citizen is chosen randomly from this group, find
- P (the citizen suffers from diabetes).
 - P (the citizen is a male and suffers from diabetes).
 - P (the citizen suffers from diabetes | female).
 - Are the two events “being a male citizen” and “suffering from diabetes” independent? Justify.
2. a) A discrete random variable X follows the following probability mass function (pmf).
- $$P(x) = \begin{cases} \frac{1}{2^x} & ; x = 1, 2, 3 \\ \frac{1}{2^{x-1}} & ; x = 4 \end{cases}$$
- Find $P(2 \leq x < 4)$.
 - If $g(x) = 3x + 2$, find $E[g(x)]$ and $Var[g(x)]$.
- b) A continuous random variable X follows the following probability density function (pdf).
- $$f(x) = \begin{cases} 5e^{-5x} & ; x > 0 \\ 0 & ; o/w \end{cases}$$
- Show that $f(x)$ is a proper pdf.
 - Derive the Moment Generation Function (MGF) of X .
 - Calculate the mean and the variance using MGF.



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
ST/MT 102 – INTRODUCTION TO PROBABILITY THEORY
MID SEMESTER EXAMINATION - AUGUST 2015

Time Allowed: ONE hour

Answer All Questions.

1. a) A stack contains 10 text books. There are 5 Statistics books, 3 Mathematics books and 2 Physics books. If you are allowed to choose only 5 books out of the 10, what is the probability that you choose
- 3 Statistics books and 2 Mathematics books.
 - At least one from each subject.
 - More Mathematics books than Statistics books.
- b) In a group of 400 citizens, 175 are males. Out of them 100 suffer from diabetics. Among the females, 200 suffer from diabetics. If a citizen is chosen randomly from this group, find
- P (the citizen suffers from diabetics).
 - P (the citizen is a male and suffers from diabetics).
 - P (the citizen suffers from diabetics | female).
 - Are the two events "being a male citizen" and "suffering from diabetics" independent? Justify.
2. a) A discrete random variable X follows the following probability mass function (pmf).
can not integrate
- $$P(x) = \begin{cases} \frac{1}{2^x} & ; x = 1, 2, 3 \\ \frac{1}{2^{x-1}} & ; x = 4 \end{cases}$$
- Find $P(2 \leq x < 4)$.
 - If $g(x) = 3x + 2$, find $E[g(x)]$ and $Var[g(x)]$.
- b) A continuous random variable X follows the following probability density function (pdf).
- $$f(x) = \begin{cases} 5e^{-5x} & ; x > 0 \\ 0 & ; o/w \end{cases}$$
- Show that $f(x)$ is a proper pdf.
 - Derive the Moment Generation Function (MGF) of X .
 - Calculate the mean and the variance using MGF.



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS & COMPUTER SCIENCE
QUIZ – 1 (SEMESTER II) – 2017/18
ST 104 – Statistical Applications II

Answer ALL Questions

Time: 75 minutes

Read the instructions carefully before you begin the exam.

You are advised to save your work regularly to avoid loss of data due to power failures.

Instructions: You should prepare a text document (Open Office Writer) including all the necessary outputs, commands and relevant interpretation. Save the file with the name XXXXXX.odt; where XXXXXX represents your registration number.

(E.g.: If the registration number is: "S/16/001" then file name should "S16001.odt")

Also save the R script file with the name "XXXXXX.R" in your home directory. XXXXXX represents your registration number.

- ✓ Save necessary files in the folder ST104_XXXXXX
- ✓ Convert your folder into a zip file for submission.
- ✓ Upload the ST104_XXXXXX.zip file to your google classroom.

1. (a). Suppose $x_0=1$, $x_1=2$ and $x_j = x_{j-1} + \frac{2}{x_{j-1}}$ for $j=1,2,\dots$

Write a function 'testLoop', which takes the single argument n and returns the first n-1 values of sequence $\{x_j\}$ $j \geq 0$: that means the values of $x_0, x_1, x_2, \dots, x_{n-2}$ (argument n should be greater than 4)

Example when using the function:

```
> testLoop(6)
[1] 1.000000 2.000000 3.000000 3.666667 4.212121
```

- (b). Now write a function 'testLoop2' which takes a single argument yVec which is a vector. The function should return

$$\sum_{j=1}^n e^j$$

where n is the length of yVec. (Hint: $\exp(1) = e^1$)

Example when using the function:

```
> testLoop2(c(2,3,4,5,8))
[1] 233.2042 3211.444
```

2. Suppose a data frame has the following structure

	Sex	Bwt	Hwt
1	F	2.0	7.0
2	F	2.0	7.4
3	F	2.0	9.5
4	F	2.1	7.2
5	F	2.1	7.3
6	F	2.1	7.6

Implement the following functions

- A function "sd_var" to find the standard deviation and variance of each numeric column. (can use inbuilt functions)
- A function "mean_median" to find the median and mean of each numeric column. (can use inbuilt functions)
- A function "correlation" to find the correlation among the numeric columns. (without using corr () function)

$$\text{Hint: } r = \frac{1}{n-1} \times \sum \left(\frac{x-\bar{x}}{s_x} \right) \left(\frac{y-\bar{y}}{s_y} \right)$$

- A function "find_table" to find the counts of the character column. *new sum*
- Create a pie chart for the categorical variable of the above data set. *each category in the*
- Use the dataset "cats" from the package "MASS" and apply your functions from (a),(b),(c) and (d). *counts of each category in the*

Example when using the functions:

```
> sd_var(MASS::cats)
The standard deviation of column 2 is : 0.4853066
The standard deviation of column 3 is : 2.434636
The variance of column 2 is : 0.2355225
The variance of column 3 is : 5.927451

> mean_median(MASS::cats)
The mean of column 2 is : 2.723611
The mean of column 3 is : 10.63056
The median of column 2 is : 2.7
The median of column 3 is : 10.1

> correlation(MASS::cats)
The correlation of column 2 and column 3 is : 0.8041274

> find_table(MASS::cats)
  F  M
47 97
```

3. "RProChallenge" is a simple programming challenge often used at interviews to test very basic programming skill. Your goal is the following: for the numbers 1 to 100,

If the number is an even print "R"

Else(if it is odd)

print "Pro" if the number is a multiple of 3,

print "Challenge" if the number is a multiple of 5,

print "ProChallenge" if the number is a multiple of both 3 and 5,

and simply print the number otherwise.

4. Create a function "Even_Odd" to count the even numbers and odd numbers of a vector. Use the function to count even and odd numbers from the following vector.

A=(2,7,4,3,1,9,6,8,5)



5/16/414

UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
ST/MT 102 – Introduction to Probability Theory
Class Test I – August 2018

Time Allowed: 45 minutes

Answer All Questions.

Select the most suitable answer among the four choices or provide your own solution.

- 1) How many five digit passcodes can be generated using integers 1,2,...9 if no digit can appear more than once?

a) 15120 b) 59049 c) 30240 d) 65610

a

- 2) Twenty workers are to be assigned to 20 different jobs, one to each job. How many different assignments are possible?

a) 19! b) 20! c) $\binom{20}{2}$ d) $\binom{19}{2}$

b

- 3) How many eight character passwords are possible if it can hold digits from 0 to 9 and letters A-Z where repetitions are allowed but cannot start with zero?

a) 36^8 b) $36!$ c) $35(36^7)$ d) $35!$

c

- 4) How many different letter arrangements can be made from the letters MISSISSIPPI?

a) $\frac{11!}{2!4!}$ b) $\frac{11!}{2!4!2!2!}$ c) $\frac{11!}{2!4!4!}$ d) $\frac{11!}{2!2!2!}$

c

- 5) If 12 people are to be divided into 3 committees of respective sizes 3, 4 and 5, how many divisions are possible?

a) 347 b) 1507 c) 86248800 d) 27720

d

- 6) A student is to answer 7 out of 10 questions in an exam. If she must answer 3 out of the first five, how many choices are there?

a) 10 b) 120 c) 50 d) 15

b

answer = 110

- 7) From a group of 8 women and 6 men, a committee consisting of 3 men and 3 women is to be formed. How many different committees are possible if 2 of the men refuse to serve together?

a) 672

b) 896

c) 336

d) 224

d

- 8) Five delegates from Russia, France, England, Germany and America are to be seated in a round table discussion. How many arrangements are possible if Russian delegate refuses to sit next to the American delegate?

a) 12

b) 24

c) 96

d) 120

a

- 9) Three girls and four boys are to be seated in a circular arrangement. How many different arrangements are possible if all three girls are always sitting together?

a) 720

b) 144

c) 48

d) 24

b

- 10) $\binom{n}{r}$ is equal to

a) $\binom{n-1}{r}$

b) $\binom{n-1}{r-1}$

c) $\binom{n-1}{r} \binom{n-1}{r-1}$

d) $\binom{n}{r-1} \binom{n-1}{r-1}$

c

- 11) $n \binom{n-1}{k-1}$ is equal to

a) $\frac{n!}{(n-k)!(k+1)!}$

b) $\frac{n!}{k!(k+1)!}$

c) $\frac{n!}{(n-k)!}$

d) $\frac{n!}{(n-k)!(k-1)!}$

d

- 12) From two groups of children where one group has n and the other group has $2n$, how many different possibilities are there if five children are to be selected from them?

a) $\binom{3n}{5}$

b) $\binom{n}{2} \binom{2n}{3}$

c) $\binom{n+3}{5}$

d) $3 \binom{n}{5}$

a

17
20

ST/MT 102 - CLASS TEST II

Reg No: 5/16/414

Choose the correct option or write your own answer.

- 1) In terms of probability, a(n) _____ is any process with uncertain results that can be repeated. 1) D
A) Event B) Outcome C) Sample space D) Experiment
- 2) The outcome of an experiment is the number of resulting heads when two coins are flipped simultaneously. What is the sample space for this experiment? 2) A
A) {HH, HT, TH, TT} B) {0, 1, 2}
C) {coin1, coin 2} D) {HH, HT, TT}
- 3) An experiment consists of randomly choosing a number between 1 and 10. Let E be the event that the number chosen is even. List the sample points in E . 3) B
A) {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} B) {1, 3, 5, 7, 9}
C) {5} D) {2, 4, 6, 8, 10}
- 4) A probability experiment is conducted in which the sample space of the experiment is $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$. Let event $A = \{5, 6, 7, 8\}$ and event $B = \{7, 8, 9, 10, 11\}$. Assume that each outcome is equally likely. List the outcomes in A and B . 4) C
A) {5, 6, 7, 8, 9, 10, 11} B) {1, 2, 3, 4, 15}
C) {7, 8} D) {1, 7, 8}
- 5) At a community college with 500 students, 120 students are age 30 or older. Find the probability that a randomly selected student is less than 30 years old. 5) A
A) .76 B) .12 C) .30 D) .24
- 6) Classify the events as dependent or independent: Events A and B where $P(A) = 0.3$, $P(B) = 0.4$, and $P(A \text{ and } B) = 0.12$. 6) B
A) dependent B) independent
- 7) You are dealt one card from a standard 52-card deck. Find the probability of being dealt an ace or a 8. 7) B
A) $\frac{13}{2}$ B) $\frac{2}{13}$ C) 9 D) $\frac{9}{26}$
- 8) Given that $P(A \text{ or } B) = \frac{1}{4}$, $P(A) = \frac{1}{5}$, and $P(A \text{ and } B) = \frac{1}{8}$, find $P(B)$. 8) B
A) $\frac{13}{40}$ B) $\frac{7}{40}$ C) $\frac{23}{40}$ D) $\frac{17}{160}$
- 9) The sample space for tossing three fair coins is {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}. What is the probability of exactly two heads? 9) D
A) 3 B) $\frac{5}{8}$ C) $\frac{1}{2}$ D) $\frac{3}{8}$

- 10) If two events have no outcomes in common they are said to be
 A) Disjoint B) Independent C) At odds D) Conditional
- 11) The events A and B are mutually exclusive. If $P(A) = 0.3$ and $P(B) = 0.5$, what is $P(A \text{ or } B)$?
 A) 0.15 B) 0.8 C) 0 D) 0.2
- 12) Suppose that E and F are two events and that $P(E \text{ and } F) = 0.36$ and $P(E) = 0.4$. What is $P(F|E)$?
 A) 0.9 B) 0.144 C) 1.111 D) 0.76
- 13) There are 32 chocolates in a box, all identically shaped. There 6 are filled with nuts, 16 with caramel, and 10 are solid chocolate. You randomly select one piece, eat it, and then select a second piece. Find the probability of selecting 2 solid chocolates in a row.
 A) $\frac{45}{512}$ B) $\frac{5}{496}$ C) $\frac{25}{256}$ D) $\frac{45}{496}$
- 14) A bag contains 10 white, 12 blue, 13 red, 7 yellow, and 8 green wooded balls. A ball is selected from the bag and kept. You then draw a second ball and keep it also. What is the probability of selecting two blue balls? Round your answer to four decimal places.
 A) 0.0588 B) 0.0539 C) 0.0528 D) 0.0576
- 15) Assume that $P(A) = 0.7$ and $P(B) = 0.2$. If A and B are independent, find $P(A \text{ and } B)$.
 A) 0.76 B) 1.00 C) 0.14 D) 0.90
- 16) A committee consisting of 6 people is to be selected from eight parents and four teachers. Find the probability of selecting three parents and three teachers.
 A) $\frac{10}{11}$ B) $\frac{100}{231}$ C) $\frac{8}{33}$ D) $\frac{2}{33}$
- 17) Suppose that B_1 and B_2 are mutually exclusive and complementary events, such that $P(B_1) = .6$ and $P(B_2) = .4$. Consider another event A such that $P(A | B_1) = .2$ and $P(A | B_2) = .5$. Find $P(B_1 | A)$.
 A) .625 B) .800 C) .240 D) .375
- 18) Use Baye's Theorem to solve this problem, A paper bag contains two red balls and one blue ball. A plastic bag contains three blue balls and one red ball. A coin is tossed. If it falls heads up, the paper bag is selected and a ball is drawn. If the coin falls tails up, the plastic bag is selected and a ball is drawn. If a red ball is selected, what is the probability that it came from the paper bag?
 A) $\frac{8}{11}$ B) $\frac{1}{8}$ C) $\frac{3}{8}$ D) $\frac{1}{3}$

TRUE/FALSE. Write 'T' if the statement is true and 'F' if the statement is false.

- 19) The expressions $A' \cap B'$ and $(A \cap B)'$ are equal.
- 20) If the probability of an event A is unaffected by the probability of an event B, then the events A and B are mutually exclusive.



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS & COMPUTER SCIENCE
END SEMESTER EXAMINATION, SEMESTER II - MARCH 2014
INTRODUCTION TO PROBABILITY - ST 102/MT102

2005
2007
2007/2008
2009
2015
2016

Time Allowed: Two hours

Answer ALL questions

Statistical Tables and Calculators will be provided.

1. A box contains 3 coins. One has two heads; one has two tails and the other is a fair coin with one head and one tail. A coin is chosen at random, is tossed and comes up heads.

- (a) What is the probability that the coin is the two-headed coin?
- (b) What is the probability that if it is tossed another time it will come up heads again?

2. Teams A and B will play 7 matches for the championship. The first team to win 4 matches will be the champion. The probability of A winning any match is 55% and there are no drawn matches. The outcome of a match does not depend on the outcome of any other matches.

- (a) What is the minimum number of matches that will have to be played? 4
- (b) If x is a number greater than or equal to your answer to part (a), what is the probability that A will win in x games?
- (c) What is the probability that A will win the championship?

(P.T.O.)

receive a scholarship?

(c) If 530 took this exam how many of them will get scholarships?

(ii) It is estimated that 80% of all 18-year old females have weights ranging from 42 to 50 kg.

Assuming that the weights are normally distributed and also that 42 and 50 are *equidistant* from the mean weight μ (i.e. same distance from μ) calculate the mean μ and the standard deviation σ for this distribution.

4. Let X be a continuous random variable whose probability density function $f_X(x)$ is given by

$$f(x) = \begin{cases} kx; & 0 \leq x < 1 \\ k(2-x); & 1 \leq x \leq 2 \\ 0; & \text{otherwise} \end{cases}$$

(a) Find the value of the constant k .

(b) Find the distribution function $F_X(x)$ for this random variable X .

(c) Find the exact value of $P(|X-1| \geq \frac{1}{2})$.

(d) A new discrete random variable Y take the value -4 when $|X-1| \geq \frac{1}{2}$ and the value 3 when $|X-1| < \frac{1}{2}$. Find the moment generating function $M_Y(t)$ of Y .



DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END-SEMESTER EXAMINATION, SEMESTER II – May 2006
STATISTICS – ST 102

Time: Two Hours

Answer ALL Questions

- ✓ 1. The probability of there being X unusable matches in a full box of a certain brand of matches is given by $P(X = 0) = 8k$, $P(X = 1) = 2k$, $P(X = 2) = P(X = 3) = k$ and $P(X \geq 4) = 0$.

Determine the constant k and the expectation and variance of X .

Two such full boxes of matches are chosen at random and the total number Y of unusable matches is determined.

- a) What are the possible values of Y ?
b) Calculate $P(Y > 4)$ and state the values of the expectation and variance of Y .

2. The mass X kg of a substance produced in one hour at a factory is modeled by a continuous random variable with pdf given by

$$f(x) = \begin{cases} kx^2; & 0 \leq x \leq 2 \\ k(6-x); & 2 \leq x \leq 6 \\ 0; & \text{otherwise} \end{cases}$$

- a) Show that $k = \frac{3}{32}$
b) Find $P(X < 4)$ and $E(X)$
c) The substance is sold at Rs 100 per kg and the running cost of the process is Rs 20 per hour. Taking Y as the profit made in one hour express Y in terms of X and determine the expected profit per hour.

3. The masses of packets of sugar are normally distributed. In a large shipment of packets of sugar, it is found that 5 % of these have a mass greater than 510 g and 2 % have a mass greater than 515 g.

- a) Determine the mean and the standard deviation of this distribution.
b) Find the probability that a randomly chosen packet will have a mass between 480 and 500 g.
c) Five of these packets are selected at random. Find the probability that the total mass exceeds 2.5 kg.

P.T.O.

5/3 0.5

4. (i) A certain variety of flower seeds is sold in packets containing 1000 seeds. It is known that 40 % of the seeds will yield white flowers and 60 % will bloom red.

a) If 5 seeds are planted, write down a formula for the probability that at least one will yield a white flower and evaluate this probability.

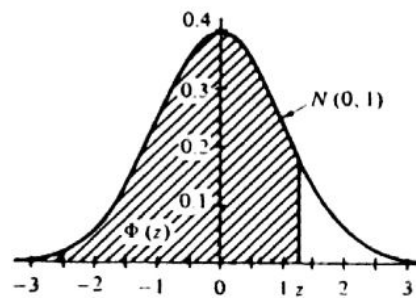
b) 100 seeds are planted. Use the normal approximation to estimate the probability of obtaining between 30 and 45 (inclusive) white flowers.

(ii)

It is known that the number of items produced at a factory during a week is a random variable with mean 50 and standard deviation 5. Place a lower bound on the probability that this week's production will be between 40 and 60.

(Hint: Use Chebyshev's inequality.)

The Normal Distribution



$$P(Z \leq z) = \Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-w^2/2} dw$$

$$[\Phi(-z) = 1 - \Phi(z)]$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990



FACULTY OF SCIENCE - UNIVERSITY OF PERADENIYA
END SEMESTER EXAMINATION - SEMESTER II, 2010(JUNE)
INTRODUCTION TO PROBABILITY THEORY - ST102/MT102 (3 CREDITS)

Time Allowed: Two hours
 Calculators will be provided

Answer All questions

1. (a). (i) How many terms will be included in the expansion of,
 $(a+b+c)(d+e+f)(x+y+u+v+w)$ 45
- (ii). Consider a population having N people of which M are adults. Assuming that samples are taken with replacement from this population, show that the probability of choosing x number of adults in a sample of size n is

$$\frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

- (b). (i). Two events A and B are defined such that;
- (i). the probability that A occurs but B does not occur is 0.2,
 - (j). the probability that B occurs but A does not occur is 0.1, and
 - (k). the probability that neither occurs is 0.6

What is $P(A|B)$? $P(A \cap B) = 0.2$
 $P(B|A) = 0.1$
 $(A \cup B) = 0.6$

- (ii). A toy manufacturer buys ball bearings from three different suppliers; 50% of his total order comes from *supplier 1*, 30% from *supplier 2*, and the rest from the *supplier 3*. Past experience has shown that the quality control standards of the three suppliers are not all the same. Two percent of the ball bearings produced by *supplier 1* are defective, while *suppliers 2* and *3* produce defective bearings 3% and 4% of the time, respectively. What proportion of the ball bearings in toy manufacturer's inventory is defective?

2. (a). (i). Define what is *probability distribution function* of a random variable X ?
dis $\rightarrow f(n) = P(X=n)$
- (ii). Based on the three axioms of probability, clearly state, the two conditions that should be satisfied by a *probability mass function*?

- (b). If X has the distribution function

$$F(x) = \begin{cases} 0 & \text{for } x < -1 \\ \frac{1}{4} & \text{for } -1 \leq x < 1 \\ \frac{1}{2} & \text{for } 1 \leq x < 3 \\ \frac{3}{4} & \text{for } 3 \leq x < 5 \\ 1 & \text{for } x \geq 5 \end{cases}$$

where $x = \dots, -2, -1, 0, 1, 2, \dots$

find,

(i). $P(X \leq 3)$

(ii). $P(X \geq 1)$

(iii). $P(-0.4 \leq X \leq 4)$

(iv). $P(X = 3)$ [P.T.O]

[Continued]

2. [Continued]

- (c). Find the expected value of the discrete random variable X having the probability distribution

$$f(x) = \frac{|x-2|}{7} \quad \text{for } x = -1, 0, 1, 3$$

3. Identify suitable distribution (or density) functions for the following scenarios, and then give suitable solutions for the given questions.

- (a). If 40% of the mice used in an experiment will become very aggressive within one minute after having been administered an experimental drug,
 (i). Find the probability that exactly six of 15 mice that have been administered the drug will become very aggressive within one minute?
 (ii). What is the expected number of mice, become very aggressive within one minute?

- (b). It is known from experience that 1.4 percent of the calls received by a reception counter are wrong numbers.

- (i). Briefly explain, under what conditions a binomial distribution can be approximated by a Poisson distribution?
 (ii). Use the Poisson approximation to the Binomial distribution to determine the probability that among 150 calls received by the counter, two are wrong numbers.

- (c). A shipment contains 100 items and a sample of size $n = 2$ is inspected. If the manufacturer decides to return all 100 items if there is at least 1 defective item in the sample, what is the probability that a shipment with 10% defective items, will be accepted?

- (d). The average amount of time that a watch will run without having to be rest is 120 days. Find the probabilities that such a watch will

- (i). have to be rest in less than 24 days;
 (ii). not have to be rest in at least 180 days

4. (a). Using moment-generating function of the binomial distribution, $M_X(t) = [1 + \theta(e^t - 1)]^n$, derive the mean and the variance of the binomial distribution.

- (b). Suppose that during periods of meditation the reduction of person's oxygen consumption is a random variable having a normal distribution with $\mu = 37.6$ cc per minute and $\sigma = 4.6$ cc per minute. Find the probabilities that during a period of meditation a person's oxygen consumption will be reduced by;

- (i). at least 44.5 cc per minute
 (ii). at most 35.0 cc per minute
 (iii). anywhere from 30.0 to 40.0 cc per minute

- (c). A biologist studying the behavior of sheep in a large field. The field is divided in to number of equally sized squares. The average number of sheep per square is 20 and the sheep are scattered randomly throughout the field.

- (i). Suggest a suitable probability model for the number of sheep in a square.
 (ii). Give a suitable value/s for any parameter/s used for the suggested model for part (i).
 (iii). Using a suitable approximation, find the probability that a randomly selected square contains fewer than 15 sheep.



FACULTY OF SCIENCE - UNIVERSITY OF PERADENIYA
END SEMESTER EXAMINATION - SEMESTER II, 2011(SEPT)
INTRODUCTION TO PROBABILITY THEORY - ST102/MT102 (3 CREDITS)

Time Allowed: Two hours
 Calculators will be provided

Answer All questions



- (a) An operation consists of two steps, of which the first can be made in n_1 ways. If the first step is made in the i th way, the second step can be made in n_{2i} ways.
- (i) Use a tree diagram to find a formula for the total number of ways in which the total operations can be made.
 - (ii) A student can study 0, 1, 2 or 3 hours for a *Statistics* test on any given day. Use the formula obtained in part (i) to verify that there are 13 ways in which the student can study at most 4 hours for the test on two consecutive day.
- (b). Assume that A , B , and C are any two events in a sample space S such that $P(A \cap B) \neq 0$.
- (i). Assuming that the three events are **NOT** independent, clearly write an expression to calculate the probability $P(A \cap B \cap C)$.
 - (ii). Prove the expression given in (b-i).
 - (iii). By using the proof given in (b-ii) derive an expression to calculate the probability $P(A \cap B \cap C)$ assuming that the three events are independent.
 - (iv). A coin is tossed three times and the eight possible outcomes, HHH, HHT, HTH, THH, HTT, THT, TTH, and TTT, are assumed to be equally likely. If A is the event that a head occurs on each of the first two tosses, B is the event that a tail occurs on the third toss, and C is the event that exactly two tails occur in the three tosses, show that,
 - (p). events A and B are independent;
 - (q). events B and C are dependent.

- 2/ (a). (i). Based on the three axioms of probability, clearly state, the two conditions that should be satisfied by a *probability mass function*.
- (ii). For the function, $f(x) = c \left(\frac{5}{x}\right)$, determine c so that the function can serve as probability distribution function of the random variable X . ($x = 1, 2, 3, 4, 5$)
- (iii). Find the distribution function of the random variable X (given in 2-a-ii).

- (b). The probability density of a random variable Y is given by

$$f(y) = \begin{cases} \frac{1}{8}(y+1), & \text{for } 2 < y < 4 \\ 0 & \text{elsewhere} \end{cases}$$

Find $P(Y < 3.2)$ and $P(2.9 < Y < 3.2)$

[P.T.O.]

[Continued]

adults - 2000000

guarantee - 2000000

- 3 Identify suitable distribution (or density) functions for the following scenarios and then give suitable solutions for the following questions.

* (a) 60% of the people in a certain population are adults. A random sample of size 15 will be drawn, with replacement, from this population.

- (i) What is the most likely number of adults in the sample?
(ii) What is the chance of getting exactly this many adults?

(b) If the probabilities of having a male or female child are both 0.50, find the probabilities that,

- (i) A family's fourth child is their first son;
(ii) A family's seventh child is their second daughter.

(c) A shipment of 80 burglar alarms contains four that are defective. If three from the shipment are randomly selected and shipped to a customer, find the probability that the customer will get exactly one bad unit.

* (d) A manufacturer guarantees that the lifetime of his tire is 40,000 kilometers. Find the probability that one of this tires will last at least 20,000 kilometers.

4 (a) In a photographic process, the developing time of prints may be looked upon as a random variable having the normal distribution with $\mu = 15.40$ and $\sigma = 0.48$ seconds. Find the probabilities that the time it takes to develop one of the prints will be

- (i) At least 15.9 seconds
(ii) At most 14 seconds
(iii) Anywhere from 15 to 16.2

(b) Explain, under which conditions the binomial distribution can be approximated using the normal distribution.

(c) If 23 percent of all patients with high blood pressure have had side effects from a certain kind of medicine, use the normal approximation to find the probability that among 120 patients with high blood pressure treated with this medicine more than 32 will have bad side effects.

0.8 - 0.23
= 0.57
9 (0.57)
= 1.0000
- 0.9982
= 0.0018

1.0000
0.9982
0.0018

1 - 0.23



8/10/818
13th Dec 2012

FACULTY OF SCIENCE - UNIVERSITY OF PERADENIYA
END SEMESTER EXAMINATION - SEMESTER II, 2012(DEC)
INTRODUCTION TO PROBABILITY THEORY – ST102/MT102 (3 CREDITS)

Time Allowed: 2 1/2 hours
Calculators will be provided

Answer All questions

1. One box contains 5 red and 6 black marbles. A second box contains 10 red and 5 black marbles. One marble is drawn from box 1 and placed in box 2. Two marbles are then drawn from box 2 without replacement.
- Draw a probability tree diagram to describe this experiment.
 - Let X be the number of red marbles drawn from box 2.
 - What are the possible values of X ?
 - Find the probability distribution of X .
 - If 2 red marbles are drawn from box 2 what is the probability that the first marble drawn from box 1 is also red?

2. Let X be a continuous random variable whose probability density function $f(x)$ is given by

$$f(x) = kx, 0 \leq x \leq 2; \quad k(4-x), 2 < x \leq 4; \quad \text{and } 0 \text{ otherwise}$$

- Find the value of k to make $f(x)$ a valid probability density function.
- Find the cumulative distribution function $F_X(t) = P(X \leq t)$ for all possible values of t and hence find the median of X .

3. A credit card company claims that their card is used by 40% of the people who buy groceries in a particular city. A random sample of 20 receipts was selected. If the company's claim is correct, use the binomial table to find the exact probability that

- from 6 to 12 (inclusive) in the sample use the credit card.
- fewer than 4 people in the sample use the card.

Use the normal approximation with continuity correction to find the approximate probabilities to parts a) and b). ($E(X) = np$; $\text{Var}(X) = np(1-p)$).

4. a) The hardness of a certain alloy is normally distributed with mean 70 units and standard deviation 3 units. A specimen is acceptable only if its hardness is between 66 and 74 units. Find the probability that a randomly chosen specimen is acceptable.

b) An instructor assigns grades based on a curve by assuming that the grades on a test are normally distributed. If the mean is 70 and the standard deviation is 8 find the test marks for each of the grades if the instructor assigns grades as follows:

Top 10% A's, next 20% B's, next 40% C's, next 20% D's and the last 10% F's.

[P.T.O]

5. a) The probability density function $f(x)$ of a random variable X is given by

$$f(x) = \frac{1}{\theta} \exp\left\{-\frac{|x - \mu|}{\theta}\right\}; -\infty < x < \infty, \mu \in R, \theta > 0.$$

Show that the moment generating function $M_X(t) = E(e^{tx})$ is $\frac{e^{\mu t}}{1 - t^2 \theta^2}; t < \frac{1}{\theta}$.

b) The mean and the variance of a random variable X are $\alpha\beta$ and $\alpha\beta^2$ respectively where $\alpha > 1$. Use Chebyshev's inequality to find a lower bound for $P(0 < X < 2\alpha\beta)$.

(Chebyshev's Inequality: $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$)

Table 1 Binomial Distribution

$$P(X \leq r) = \sum_{i=0}^r \binom{n}{i} p^i (1-p)^{n-i}$$

		<i>p</i>									
<i>n</i>	<i>r</i>	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
20	0	0.3585	0.1216	0.0388	0.0115	0.0032	0.0008	0.0002	0.0000	0.0000	0.0000
	1	0.7358	0.3917	0.1756	0.0692	0.0243	0.0076	0.0021	0.0005	0.0001	0.0000
	2	0.9245	0.6769	0.4049	0.2061	0.0913	0.0355	0.0121	0.0036	0.0009	0.0002
	3	0.9841	0.8670	0.6477	0.4114	0.2252	0.1071	0.0444	0.0160	0.0049	0.0013
	4	0.9974	0.9568	0.8298	0.6296	0.4148	0.2375	0.1182	0.0510	0.0189	0.0059
	5	0.9997	0.9887	0.9327	0.8042	0.6172	0.4164	0.2454	0.1256	0.0553	0.0207
	6	1.0000	0.9976	0.9781	0.9133	0.7858	0.6080	0.4166	0.2500	0.1299	0.0577
	7	1.0000	0.9996	0.9941	0.9679	0.8982	0.7723	0.6010	0.4159	0.2520	0.1316
	8	1.0000	0.9999	0.9987	0.9900	0.9591	0.8867	0.7624	0.5956	0.4143	0.2517
	9	1.0000	1.0000	0.9998	0.9974	0.9861	0.9520	0.8782	0.7553	0.5914	0.4119
	10	1.0000	1.0000	1.0000	0.9994	0.9961	0.9829	0.9468	0.8725	0.7507	0.5881
	11	1.0000	1.0000	1.0000	0.9999	0.9991	0.9949	0.9804	0.9435	0.8692	0.7483
	12	1.0000	1.0000	1.0000	1.0000	0.9998	0.9987	0.9940	0.9799	0.9420	0.8684
	13	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9985	0.9935	0.9786	0.9423
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9984	0.9936	0.9793
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9985	0.9941
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9987
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION, SEMESTER II – DECEMBER, 2018
ST/MT102 – INTRODUCTION TO PROBABILITY THEORY – PART I.

Time Allowed: **TWO Hours**

Answer **All Questions.**

1.

- a). Show that the number of ways in which a set of n distinct objects can be partitioned into k subsets with n_1 objects in the first subset, n_2 objects in the second subset,, and n_k objects in the k^{th} subset is,

$$\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! \cdot n_2! \cdot \dots \cdot n_k!}$$

- b). How many 5-digit numbers can be formed from the integers, 1,2,...,9 if no digit can be appeared more than twice?
- c). If n balls are placed at random into n cells, find the probability that exactly one cell remains empty.

[Hint: If exactly one cell is empty, there will be exactly one cell having two balls]

2. The probability density function (pdf) of a random variable X , the life time of a certain type of electronic device (measured in hours), is given by

$$f(x) = \begin{cases} \frac{10}{x^2} & ; \quad x > 10 \\ 0 & ; \quad x \leq 10 \end{cases}$$

- a). Show that $f(x)$ is a legitimate pdf.
- b). Derive the Cumulative Density Function (CDF) of X .
- c). Calculate $P(x > 20)$.
- d). What is the probability that this electronic device will have more than 30 hours of life time given that it already had more than 20 hours of life?
- e). What is the probability that of 6 such type of devices, exactly 3 will function for at least 20 hours? State the assumptions you make.

[P.T.O.]

3.

a). Let X be a continuous random variable where the moment generating function, $M_x(t)$ exists.

For constants a and b , show that in usual notation,

$$M_{\frac{(x+a)}{b}}(t) = e^{\left(\frac{a}{b}\right)t} M_x(t/b)$$

b). Suppose that a random variable X has the geometric distribution with probability of success θ .

(i) Show that the moment generating function (MGF) of X is

$$M_x(t) = \frac{\theta e^t}{1 - (1 - \theta)e^t}$$

(ii) Use the MGF to show that $E(X) = 1/\theta$.

(iii) A survey was conducted on customers leaving a service station where the satisfaction rate in a usual day is known to be 75%. What is the probability that the first satisfied customer who leaves the station is the 10th customer of the day?

4. Let a continuous random variable X follows uniform distribution within the interval $(0,10)$.

a). Use Markov's inequality to find an upper bound for $P(X > 8)$

b). Use Chebyshev's inequality to find a lower bound for the probability of X within two standard deviations of mean.

c). Calculate the exact probability for part(b).

d). If Y is a normal random variable with mean μ and standard deviation σ , find $P(|y - \mu| > 2\sigma)$.

Distribution	Density Function
Binomial (n, θ)	$P(X = x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}; x = 0, 1, 2, \dots, n$
Poisson (λ)	$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, \dots$
Negative Binomial (k, θ)	$P(X = x) = \binom{x-1}{k-1} \theta^k (1 - \theta)^{x-k}; x = k, k+1, k+2, \dots$
Geometric (θ)	$P(X = x) = \theta (1 - \theta)^{x-1}; x = 1, 2, \dots$
Uniform(a, b)	$f(x) = \frac{1}{b-a}; a \leq x \leq b$